Physics of rubber — Solution

Y26 (2026) — English translation

A1 **0.40 pts**

Find the number of microscopic states Γ, with the help of which the macroscopic state is realized, in which the difference in the number of links oriented along and against the axis x is equal to $m = n_1 - n_2$. Also obtain an approximate answer for $m \ll n$. Express your answers using m and n.

For a given m, the number of links oriented in the positive direction is $n_1 = \dfrac{n+m}{2}$, and in the negative direction $n_2 = \dfrac{n-m}{2}$. Then the required number of ways

Answer: \textbfAnswer:

$$\Gamma = \frac{n!}{\left(\frac{n+m}{2}\right)!\left(\frac{n-m}{2}\right)!} \approx 2^n \sqrt{\frac{2}{\pi n}} \exp\left(-\frac{m^2}{2n}\right)$$

A2 **0.80 pts**

Show that at $h_x \ll na$ the entropy of the molecule has the form

$$S = S_0 - \gamma h_x^2.$$

Express coefficient γ via/through n, a and k.

The displacement of the end of the molecule is $h_x = (n_1 - n_2)a = ma$. Then, using the formula from the previous paragraph, we get

$$\Gamma = 2^n \sqrt{\frac{2}{\pi n}} \exp\left(-\frac{h_x^2}{2na^2}\right).$$

Then from the definition энтропии we get

$$S = k \ln \Gamma = k \ln\left(2^n \sqrt{\frac{2}{\pi n}}\right) - \frac{k h_x^2}{2na^2}.$$

Answer: \textbfAnswer:

$$\gamma = \frac{k}{2na^2}$$

A3 **0.60 pts**

The probability that the end of a molecule lies in the interval $[h_x, h_x + dh_x]$ of width $dh_x \gg a$ is $f(h_x)dh_x$. Show that for $|h_x| \ll na$ the probability density of $f(h_x)$ has the form

$$f(h_x) = A \exp(-\alpha h_x^2)$$

and вытimesandme постоянные A and α via/through n and a. Считайте, что направления звеньев цепи независимы друг от друга and equalsвероятны. For/Atмечание: for/at повороте одного звена h_x меняется на $2a$.

Note: When one link is rotated, h_x changes to $2a$.

The total number of states in the circuit is 2^n. Since the directions of the chain links are equally probable, the probability of a given state of the molecule is proportional to the number of microscopic states in which it can be realized. Then the probability that the displacement of the end of the molecule takes on a given value h_x is equal to

$$p(h_x) = \frac{1}{2^n}\Gamma(h_x) = \sqrt{\frac{2}{\pi n}}\exp\left(-\frac{h_x^2}{2na^2}\right).$$

For/At повороте одного звена value m меняется на 2, therefore смещение h_x меняется на $2a$. Значит в некотором интервале ширины $dh_x \gg a$ находится $dN = dh_x/2a$ возможных состояний. For/At достаточно малом dh_x вероятности этих состояний практически одинаковы, and therefore вероятность попадания в данный интервал equals

$$dN \cdot p(h_x) = \frac{dh_x}{2a}\sqrt{\frac{2}{\pi n}}\exp\left(-\frac{h_x^2}{2na^2}\right).$$

Hence density вероятности

$$f(h_x) = \frac{1}{a\sqrt{2\pi n}}\exp\left(-\frac{h_x^2}{2na^2}\right).$$

Taking into account expression for гауссова интеграла

$$\int\limits_{-\infty}^{+\infty} dx\, e^{-\beta x^2} = \sqrt{\frac{\pi}{\beta}}$$

we get

$$\int\limits_{-\infty}^{+\infty} dh_x\, f(h_x) = \frac{1}{a\sqrt{2\pi n}}\sqrt{2\pi n a^2} = 1,$$

that is, the distribution is correctly normalized.

Answer: \textbfAnswer:

$$A = \frac{1}{a\sqrt{2\pi n}}, \quad \alpha = \frac{1}{2na^2}$$

B1 **0.50 pts**

What are the values of $\langle h^2\rangle$ and $\langle h_x^2\rangle$ for a three-dimensional chain? Express your answers using n and l. Note: if you were unable to complete this step, you can continue to count $\langle h_x^2\rangle = nl^2$ throughout the entire task.

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The displacement vector of the end of the vector relative to the beginning is equal to the sum of the vectors $\vec{l}_i$,

$$\vec{h} = \sum_i \vec{l}_i,$$

therefore его квадрат

$$\vec{h}^2 = \left(\sum_i \vec{l}_i\right)^2 = \sum_i (\vec{l}_i)^2 + 2\sum_{i<j}\vec{l}_i\vec{l}_j.$$

Here в первой сумме суммирование производится по всем звеньям молекулы, а во второй – по всем vaporам звеньев. Каждое term первой суммы equals длине одного звена l^2, therefore вся первая сумма equals nl^2. Усредним вторую сумму по сличным состояниям молекулы. В силу независимости направлений сличных звеньев каждое from произведений for/at усреднении gives 0, $\langle \vec{l}_i \vec{l}_j \rangle = 0$, therefore and вся сумма for/at усреднении обращается в 0. Then for среднего квадрата we get

$$\langle h^2 \rangle = nl^2.$$

В силу equalsвероятности всех направлений

$$\langle h_x^2 \rangle = \langle h_y^2 \rangle = \langle h_z^2 \rangle.$$

Since

$$\langle h^2 \rangle = \langle h_x^2 \rangle + \langle h_y^2 \rangle + \langle h_z^2 \rangle,$$

средний квадрат одной from проекций equals

$$\langle h_x^2 \rangle = \frac{1}{3}\langle h^2 \rangle = \frac{1}{3} nl^2.$$

Answer: \textbfAnswer:

$$\langle h^2 \rangle = nl^2, \quad \langle h_x^2 \rangle = \frac{1}{3} nl^2$$

B2 1.00 pts

What is the average distance between the ends of the chain $\langle h \rangle$? Express your answer in terms of n and l.

Differentiating the value of the Gaussian integral

$$\int\limits_{-\infty}^{+\infty} dx\, e^{-\beta x^2} = \sqrt{\frac{\pi}{\beta}}$$

по parametery β we get

$$\int\limits_{-\infty}^{+\infty} dx\, x^2 e^{-\beta x^2} = \frac{1}{2}\sqrt{\frac{\pi}{\beta^3}}, \quad \int\limits_{-\infty}^{+\infty} dx\, x^4 e^{-\beta x^2} = \frac{3}{4}\sqrt{\frac{\pi}{\beta^5}}.$$

From the condition нормировки плотности вероятности

$$\int d^3 h f(\vec{h}) = 1 = A \int d^3 h e^{-\beta h^2} = 4\pi A \int\limits_0^{\infty} h^2 dh\, e^{-\beta h^2} = \pi A \frac{\sqrt{\pi}}{\beta^{3/2}}$$

we find coefficient

$$A = \left(\frac{\beta}{\pi}\right)^{3/2}.$$

Note that for/at вычислении интеграла появился дополнительный множитель $1/2$, since нижний limit equals 0, not $-\infty$. Then for среднего квадрата we get

$$\langle h^2 \rangle = \int d^3 h\, h^2 f(\vec{h}) = 4\pi A \int\limits_0^{\infty} h^4 dh\, e^{-\beta h^2} = \pi A \frac{3}{2} \frac{\sqrt{\pi}}{\beta^{5/2}} = \frac{3}{2\beta}.$$

Using expression from предыдущего parta, we find

$$\frac{3}{2\beta} = nl^2,$$

hence

$$\beta = \frac{3}{2nl^2}.$$